LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to aircraft structural principles, general assumptions in engineering theory
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to aircraft structural principles, general assumptions in engineering theory. methods by which structure carries load, typical ideal bending movement and shear force diagram.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of structural element diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aircraft in flight?
(b) Define the term 'structure' in the context of aircraft engineering.
(c) List three common structural elements found in an aircraft fuselage.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General assumptions in engineering theory • Methods by which structure carries load • Types of structural elements and details	L & Ds	PPT, IB	15 min
(b)	• Typical ideal bending movement characteristics • Shear force diagram interpretation • Load path analysis in aircraft structure	L & Ds	PPT, PDF	20 min
(c)	• Integration of structural principles in design • Review of specific structural element functions • Summary of load carrying methods	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a typical structural element carries a bending load.
(b) Describe the relationship between shear force and bending moment in an ideal beam.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from general structural assumptions to specific fuselage design principles as covered in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fuselage introduction, fuselage shape, structural design, detailed design layout	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min